The Price of Strategyproofness for Social Welfare with Three Resources

Draft note

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Abstract

We consider the Leontief multi-resource allocation model with normalized demands, share incentive, envy freeness, Pareto optimality, and strategyproofness. Bei, Li, and Luo asked for the exact value of the price of strategyproofness for utilitarian social welfare when the number of resources is three; the previously recorded bounds were between 2 and 3. This note gives a matching lower bound of 3. Since the general upper bound is also 3, the exact value is

$$\mathrm{PoSP}_{\mathrm{SW}}(3) = 3.$$

In fact, the lower bound applies to every deterministic mechanism satisfying only share incentive and strategyproofness.

1 Model and statement

There are n agents and m divisible resources, each with unit supply. An agent i has a normalized demand vector $d_i = (d_{i1}, \dots, d_{im}) \in (0,1]^m$ with $\max_r d_{ir} = 1$. Given a bundle $X \in \mathbb{R}_+^m$, the Leontief value of X for demand d is

$$v_d(X) = \min_{r \in [m]} \frac{X_r}{d_r}.$$

Thus, under allocation A, agent i receives utility $u_i(A_i) = v_{d_i}(A_i)$.

A feasible allocation satisfies $\sum_i A_{ir} \leq 1$ for every resource r. Share incentive (SI) means that every agent receives utility at least $1/n$. Envy freeness (EF) means that $v_{d_i}(A_i) \geq v_{d_i}(A_j)$ for all agents i, j. Pareto optimality (PO) means that no feasible allocation weakly improves all agents and strictly improves at least one. Strategyproofness (SP) means that, for every true demand d_i, replacing the report of agent i by another normalized demand vector cannot increase the utility evaluated with respect to d_i.

The social welfare of an allocation is

$$\mathrm{SW}(A) = \sum_i u_i(A_i).$$

For an instance I, let

$$\mathrm{OPT}_{\mathrm{fair}}(I) = \max\{\mathrm{SW}(A) : A \text{ is feasible, SI, and EF}\}.$$

For a mechanism f, define

$$\mathrm{FR}_{\mathrm{SW}}(f) = \sup_{I} \frac{\mathrm{OPT}_{\mathsf{fair}}(I)}{\mathrm{SW}(f(I))}.$$

For a fixed number m of resources,

$$\mathrm{PoSP}_{\mathrm{SW}}(m) = \inf_{f \text{ satisfies SI, EF, PO, SP}} \mathrm{FR}_{\mathrm{SW}}(f).$$

The open question considered here is the case $m = 3$.

Theorem 1. *For deterministic mechanisms in the above model,*

$$\mathrm{PoSP}_{\mathrm{SW}}(3) = 3.$$

More strongly, every deterministic mechanism satisfying SI and SP has $\mathrm{FR}_{\mathrm{SW}}(f) = 3$ *when* $m = 3$*.*

2 The easy upper bound

Proposition 1. *For three resources, every SI mechanism has fair-ratio at most* 3*. Consequently, since standard dominant-resource-fairness mechanisms provide nonempty examples satisfying SI, EF, PO, and SP, one has* $\mathrm{PoSP}_{\mathrm{SW}}(3) \le 3$*.*

Proof. Fix any feasible allocation A and, for every agent i, choose a dominant resource $r(i)$ with $d_{i,r(i)} = 1$. Then

$$u_i(A_i) = v_{d_i}(A_i) \le A_{i,r(i)}.$$

Summing by dominant resource,

$$\mathrm{SW}(A) \le \sum_{r=1}^{3} \sum_{i:r(i)=r} A_{ir} \le 3.$$

On the other hand, if f satisfies SI, then $\mathrm{SW}(f(I)) \ge n \cdot (1/n) = 1$ on every instance I. Hence $\mathrm{OPT}_{\mathsf{fair}}(I)/\,\mathrm{SW}(f(I)) \le 3$ for every I. $\square$

It remains to prove that no SI and SP mechanism can do better.

3 A hard family of instances

Fix an integer $q \ge 3$. Later we take $q \to \infty$. Choose n much larger than q; throughout the proof it is enough to take n sufficiently large as a function of q. Let

$$n_0 = n - 2q, \qquad \eta = n^{-2}, \qquad \beta = n^2,$$

and, for $t = 1, \ldots, q$, set

$$\alpha_t = n^{-2q-4}\beta^t = n^{2t-2q-4}.$$

Thus

$$0 < \alpha_1 < \alpha_2 < \cdots < \alpha_q \le n^{-4}, \qquad \frac{\alpha_{t+1}}{\alpha_t} = \beta = n^2.$$

There are three groups of agents. The first group P contains n_0 agents, each with demand

$$a = (1, \eta, \eta).$$

The second group contains agents $B_1, \dots, B_q$ with original demands

$$b_t = \left(\frac{1}{q}, 1, \alpha_t\right), \qquad t = 1, \dots, q.$$

The third group contains agents $C_1, \dots, C_q$ with original demands

$$c_t = \left(\frac{1}{q}, \alpha_t, 1\right), \qquad t = 1, \dots, q.$$

For each t define

$$\gamma_t = \frac{q(q-1)}{tn}.$$

For n large, $0 < \gamma_t < 1$. We shall also use the lowered demands

$$\widehat{b}_t = \left(\frac{\gamma_t}{q}, 1, \gamma_t \alpha_t\right), \qquad \widehat{c}_t = \left(\frac{\gamma_t}{q}, \gamma_t \alpha_t, 1\right).$$

These are normalized demand vectors: the dominant resource remains resource 2 for $\widehat{b}_t$ and resource 3 for $\widehat{c}_t$.

Fix a deterministic mechanism f satisfying SI and SP. We first find a pair (j, k) whose two corresponding agents are resource-poor in suitable one-agent modified profiles.

3.1 A weighted pigeonhole lemma

Let

$$H_q = \sum_{t=1}^{q} \frac{1}{t}, \qquad T_t = \frac{6q}{nH_q t}.$$

For a fixed k, let $I^{\widehat{c}_k}$ denote the profile in which only C_k's demand is changed from c_k to $\widehat{c}_k$; all B-agents have their original demands. Let $R^B_{j,k}$ be the amount of resource 1 allocated by f to agent B_j on profile $I^{\widehat{c}_k}$.

Similarly, for a fixed j, let $I^{\widehat{b}_j}$ denote the profile in which only B_j's demand is changed from b_j to $\widehat{b}_j$; all C-agents have their original demands. Let $R^C_{j,k}$ be the amount of resource 1 allocated by f to agent C_k on profile $I^{\widehat{b}_j}$.

Lemma 1. *There exists a pair $(j, k) \in [q]^2$ such that*

$$R^B_{j,k} < T_j \qquad \text{and} \qquad R^C_{j,k} < T_k.$$

Proof. Consider a fixed k. On profile $I^{\widehat{c}_k}$, SI gives each of the $n_0 = n - 2q$ agents in group P utility at least $1/n$. Since their first demand coordinate is 1, these agents consume at least $(n - 2q)/n$ units of resource 1. Therefore all agents outside P together consume at most $2q/n$ units of resource 1, and in particular

$$\sum_{j=1}^{q} R^B_{j,k} \le \frac{2q}{n}.$$

Let $w_j = (H_q j)^{-1}$, so that $\sum_{j=1}^{q} w_j = 1$. Since $T_j = (6q/n)w_j$, the set

$$\mathcal{B}_k = \{j : R^B_{j,k} \geq T_j\}$$

has w-weight at most 1/3:

$$\frac{6q}{n} \sum_{j \in \mathcal{B}_k} w_j = \sum_{j \in \mathcal{B}_k} T_j \leq \sum_{j=1}^{q} R^B_{j,k} \leq \frac{2q}{n}.$$

By the same argument, for every fixed j, the set

$$\mathcal{C}_j = \{k : R^C_{j,k} \geq T_k\}$$

has w-weight at most 1/3.

Now put the product probability measure $w_j w_k$ on $[q]^2$. The set of pairs violating the first desired inequality has measure at most 1/3, and the set violating the second desired inequality has measure at most 1/3. Their union has measure at most 2/3, so some pair violates neither inequality. □

Fix from now on a pair (j, k) satisfying Lemma 1. Let $I^{j,k}$ denote the final profile obtained from the original profile by replacing b_j by $\widehat{b}_j$ and c_k by $\widehat{c}_k$, leaving all other demands unchanged.

3.2 Strategyproofness makes the chosen agents low-utility

Lemma 2. *On the final profile* $I^{j,k}$,

$$\mathrm{SW}(f(I^{j,k})) \leq 1 + \frac{2q^2}{n} + \frac{12q}{(q-1)H_q}.$$

Proof. We first bound the utility of the changed agent B_j. In profile $I^{\widehat{c}_k}$, agent B_j still has true demand b_j. By the choice of (j, k), the first-resource amount allocated to B_j is less than T_j. Since the first coordinate of b_j is $1/q$, the resulting truthful utility is less than qT_j.

Now compare with the final profile $I^{j,k}$. Suppose that, on $I^{j,k}$, agent B_j receives utility u with respect to the lowered demand $\widehat{b}_j$. If in profile $I^{\widehat{c}_k}$ the same agent, whose true demand is still b_j, misreported $\widehat{b}_j$, the mechanism would return exactly the bundle it returns to B_j on $I^{j,k}$. Any bundle worth u under $\widehat{b}_j$ is worth at least $\gamma_j u$ under b_j: the first and third coordinates of $\widehat{b}_j$ are scaled by γ_j, while the second coordinate is unchanged and $\gamma_j < 1$. Strategyproofness therefore gives

$$\gamma_j u < qT_j.$$

Consequently

$$u < \frac{qT_j}{\gamma_j} = \frac{q}{q(q-1)/(jn)} \cdot \frac{6q}{nH_q j} = \frac{6q}{(q-1)H_q}.$$

The same argument, using the profile $I^{\widehat{b}_j}$, gives the identical bound for the changed agent C_k.

It remains to bound the welfare of all other agents. The group P has total utility at most 1, because each such agent has first demand coordinate 1 and the total supply of resource 1 is 1. In the final profile, SI for the $n - 2q$ agents in P again leaves at most $2q/n$ units of resource 1 for all agents outside P. Every unchanged agent outside P has first demand coordinate $1/q$, and hence utility at most q times her first-resource amount. Their total utility is therefore at most $q(2q/n) = 2q^2/n$. Adding the two changed agents gives the claimed bound. □

4 A nearly welfare-3 fair allocation for the same profile

We now show that the final profile $I^{j,k}$ admits an SI and EF allocation with social welfare tending to 3.

Define

$$D_t = q\left(t - 1 + \frac{1}{\gamma_t}\right) = q(t-1) + \frac{tn}{q-1},$$

and set

$$x_B = \frac{1 - 3q/n}{D_j}, \qquad x_C = \frac{1 - 3q/n}{D_k}.$$

For n large enough these numbers are positive. We define an allocation A^* as follows.

- Every agent in group P receives the bundle $(1/n)a$.
- For $h < j$, agent B_h receives the bundle $(qx_B)b_h$ and hence utility qx_B.
- Agent B_j, whose final demand is $\widehat{b}_j$, receives the bundle $(qx_B/\gamma_j)\widehat{b}_j$ and hence utility qx_B/γ_j.
- For $h > j$, agent B_h receives the bundle $(1/n)b_h$.
- For $\ell < k$, agent C_ℓ receives the bundle $(qx_C)c_\ell$ and hence utility qx_C.
- Agent C_k, whose final demand is $\widehat{c}_k$, receives the bundle $(qx_C/\gamma_k)\widehat{c}_k$ and hence utility qx_C/γ_k.
- For $\ell > k$, agent C_ℓ receives the bundle $(1/n)c_\ell$.

The selected B-agents, namely $B_1, \dots, B_j$, each receive exactly x_B units of resource 1. Their total utility is

$$(j-1)qx_B + \frac{qx_B}{\gamma_j} = D_j x_B = 1 - \frac{3q}{n}.$$

Similarly, the selected C-agents have total utility $1 - 3q/n$. All remaining agents receive utility exactly $1/n$. Hence

$$\begin{aligned}\mathrm{SW}(A^*) &= \left(1 - \frac{3q}{n}\right) + \left(1 - \frac{3q}{n}\right) + \frac{(n-2q) + (q-j) + (q-k)}{n} \\ &= 3 - \frac{6q + j + k}{n} \geq 3 - \frac{8q}{n}. \end{aligned} \tag{1}$$

Lemma 3. *For each fixed q, if n is sufficiently large, then A^* is feasible.*

Proof. Resource 1 is the only resource for which the accounting is tight. Since $D_t \geq tn/(q-1)$,

$$tx_t \leq \frac{q-1}{n} \qquad (t = j, k),$$

where $x_j = x_B$ and $x_k = x_C$. Thus the selected B- and C-agents together use at most $2(q-1)/n$ units of resource 1. The agents in P use $(n-2q)/n$ units, and the unselected B- and C-agents use at most $2/n$ more. The total is at most

$$\frac{n-2q}{n} + \frac{2(q-1)}{n} + \frac{2}{n} = 1.$$

For resource 2, the selected B-agents use exactly $1 - 3q/n$. The unselected B-agents use at most q/n. The agents in P use at most $\eta = n^{-2}$. The selected C-agents use at most

$$qx_C \sum_{\ell \leq k} \alpha_\ell \leq qx_C\, k\alpha_k \leq qn^{-4},$$

for n large, because $qx_C \leq 1$ and $k \leq q$. The unselected C-agents use at most qn^{-5}. Hence resource 2 usage is at most

$$1 - \frac{3q}{n} + \frac{q}{n} + n^{-2} + qn^{-4} + qn^{-5} < 1$$

for sufficiently large n. The resource 3 calculation is symmetric. □

Lemma 4. *For each fixed q, if n is sufficiently large, then A^* is SI and EF.*

Proof. Share incentive is immediate for the agents receiving utility $1/n$. For the selected agents, it is enough to show $qx_B \geq 1/n$ and $qx_C \geq 1/n$; the changed selected agents receive even more. Since $j, k \leq q$ and $q \geq 3$, these inequalities hold for all sufficiently large n from the formula

$$qx_t = \frac{q(1 - 3q/n)}{q(t-1) + tn/(q-1)}, \qquad t \leq q.$$

We prove envy freeness. First observe that no agent with utility at least $1/n$ envies a bundle of the form $(1/n)d'$ for a normalized demand vector d': if d is the evaluating demand and r is a dominant resource for d, then $d_r = 1$ and $d'_r \leq 1$, so

$$v_d((1/n)d') \leq \frac{d'_r}{nd_r} \leq \frac{1}{n}.$$

Thus it suffices to check envy toward the selected B- and C-bundles.

Within the selected B-agents, all selected bundles contain exactly x_B units of resource 1. For a selected B-agent, the ratio of this common first-resource amount to her own first demand coordinate is exactly her own utility. Therefore she values every selected B-bundle at no more than her own utility. The same argument applies within the selected C-agents.

Next consider cross-envy between selected B- and selected C-agents. Every selected C-bundle contains at most $qx_C\alpha_k$ units of resource 2, while every selected B-agent has second demand coordinate 1. Since $\alpha_k \leq n^{-4}$ and $qx_C \leq 1$ for n large, this value is less than $1/n$, and hence less than the own utility of any selected B-agent. The symmetric argument rules out envy by selected C-agents toward selected B-bundles.

It remains to show that agents with baseline utility $1/n$ do not envy selected bundles. A group-P agent values any selected B-bundle at most its third resource amount divided by η. This is at most

$$\frac{qx_B\alpha_j}{\eta} < \frac{1}{n}$$

for n large, since $\alpha_j \leq n^{-4}$, $\eta = n^{-2}$, and $qx_B \leq 1$. The same argument, using resource 2, handles selected C-bundles.

Now take an unselected B_h with $h > j$. Such an agent values any selected B-bundle at most through the third coordinate:

$$\frac{qx_B\alpha_j}{\alpha_h} \leq \frac{qx_B}{\beta} < \frac{1}{n},$$

because $\alpha_h/\alpha_j \geq \beta = n^2$. The same agent values every selected C-bundle by at most its resource-2 amount, which is at most $qx_C\alpha_k < 1/n$. The argument for unselected C-agents is symmetric. Thus no agent envies another agent's bundle. $\square$

Combining Lemmas 3 and 4 with (1),

$$\mathrm{OPT}_{\mathrm{fair}}(I^{j,k}) \geq 3 - \frac{8q}{n}.$$

Together with Lemma 2, this gives

$$\mathrm{FR}_{\mathrm{SW}}(f) \geq \frac{3 - 8q/n}{1 + 2q^2/n + 12q/((q-1)H_q)}.$$

First choose n arbitrarily large as a function of q, and then let $q \to \infty$. Since $H_q \to \infty$, the right-hand side tends to 3. Therefore every deterministic SI and SP mechanism satisfies $\mathrm{FR}_{\mathrm{SW}}(f) \geq 3$. Proposition 1 gives the reverse inequality, so every deterministic SI and SP mechanism has fair-ratio exactly 3 for three resources. In particular,

$$\mathrm{PoSP}_{\mathrm{SW}}(3) = 3.$$

Acknowledgment

This note addresses the question formulated by Bei, Li, and Luo for the three-resource case of their price-of-strategyproofness problem.

References

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